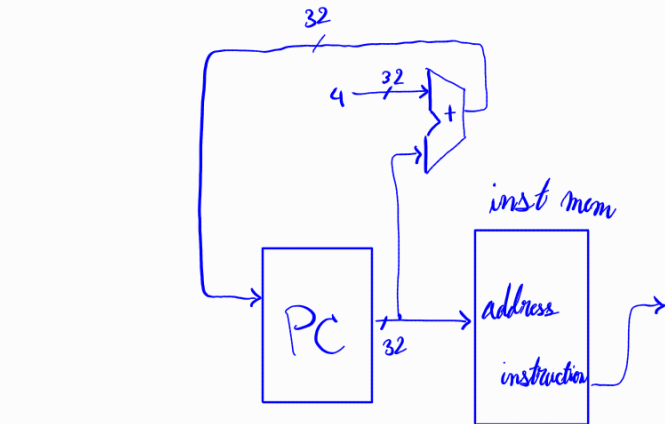


سیدمانی میر شعیبانی
۸۱۰۸۰۱۰۸۰

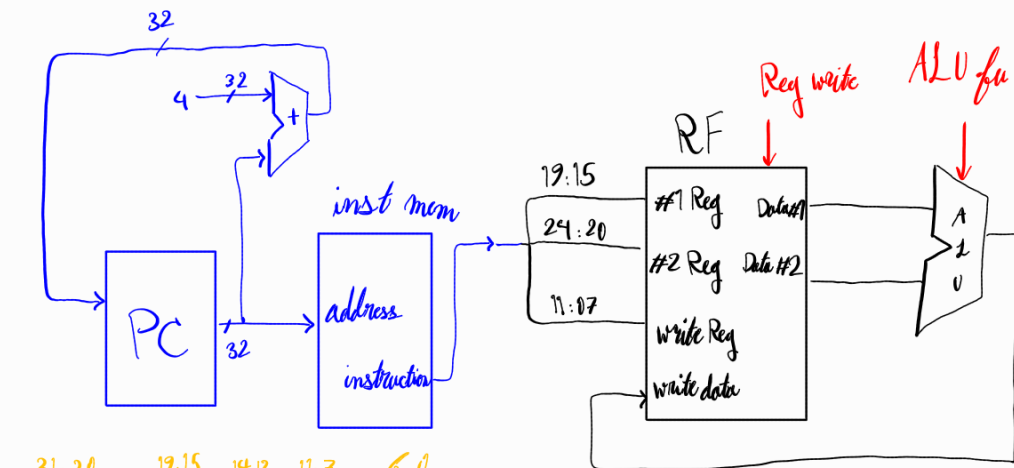
”بسم الله الرحمن الرحيم“
تمرین کا بیرونی دوم
دعائی کا بیرونی

محمد اکبری شہید
۸۱۰۸۰۱۰۸۰

رسم سر دادہ : بخش توالی (دستی)



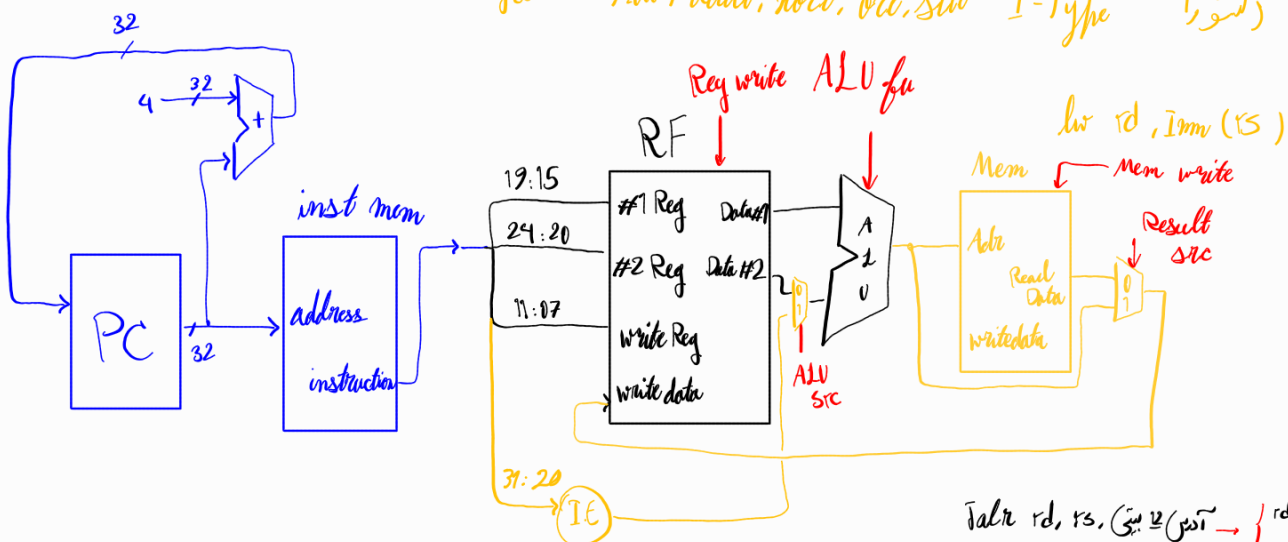
افانہ کردن دستورات R-Type :



add, sub, or, slt



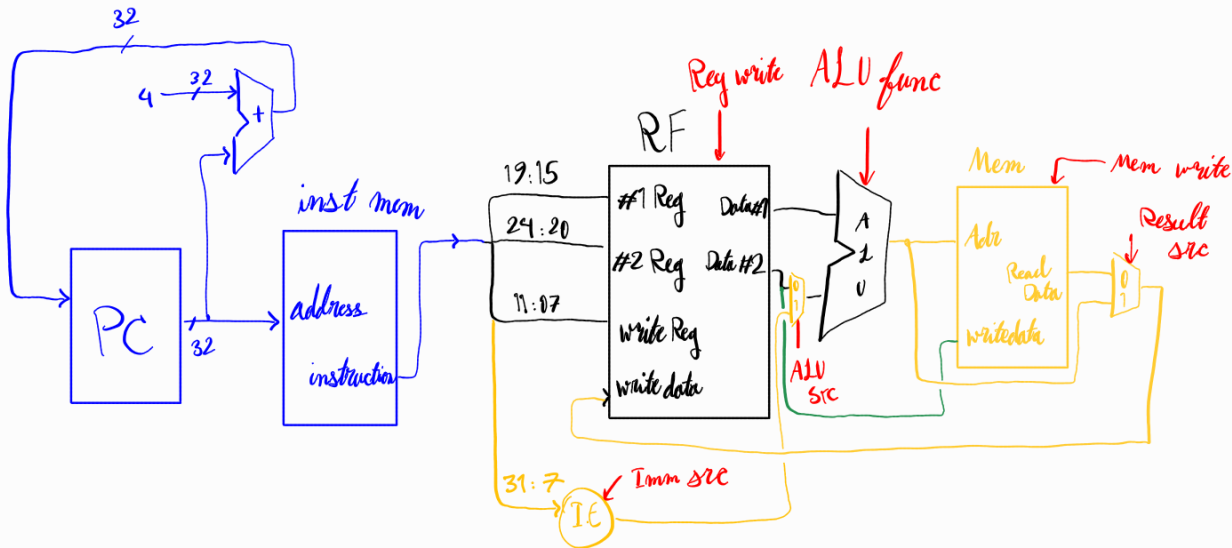
افانہ کردن دستورات I-Type : jalr, lw, addi, xori, ori, sli



jalr rd, rs, Imm (آفس ۱۲ بیتی) → $rd \leftarrow PC + 4$
 $PC \leftarrow rs + \text{آفس ۱۲ بیتی}$

31:25	24:20	19:15	14:12	11:7	6:0
Imm[19:5]	rs	rs	rd	Imm[4:0]	OP

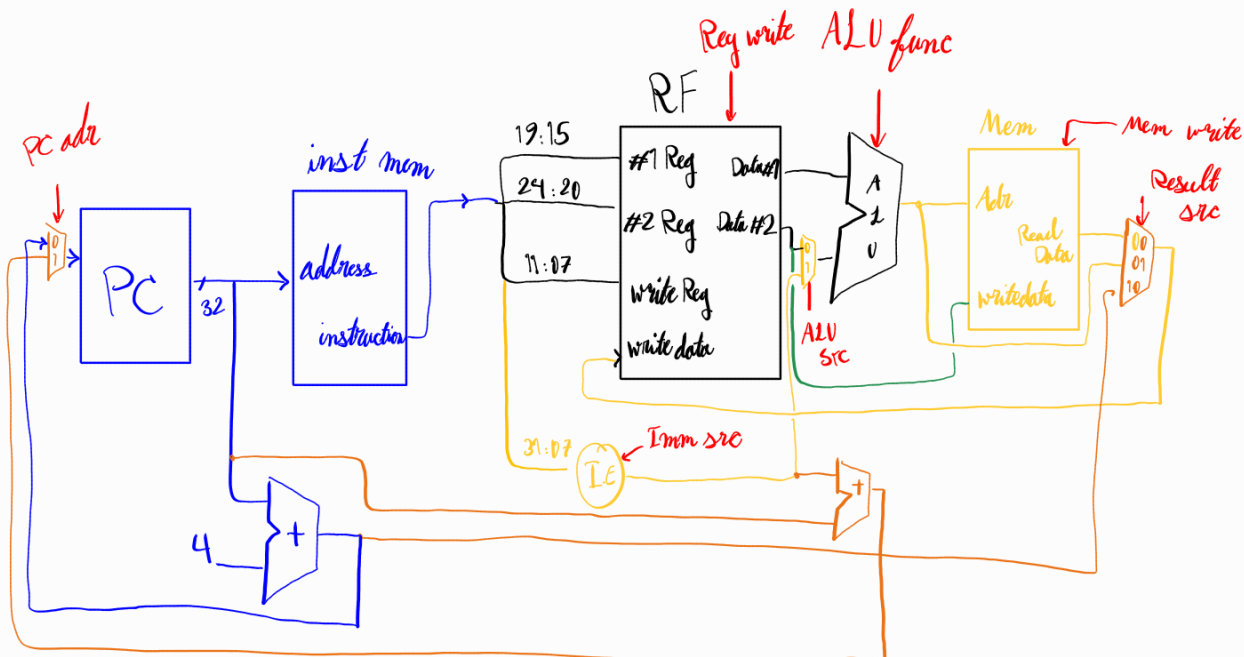
افزاد کردن دستور S-Type
SW rd, Imm(rs)



31:12	11:7	6:0
Imm[20], Imm[10:1], Imm[7], Imm[19:12]	rd	OP

افزاد کردن دستور J-Type

Jal rd, adr $\rightarrow$ $\begin{cases} rd \leftarrow PC + 4 \\ PC \leftarrow PC + adr \end{cases}$



31:20

24:20

19:15

14:12

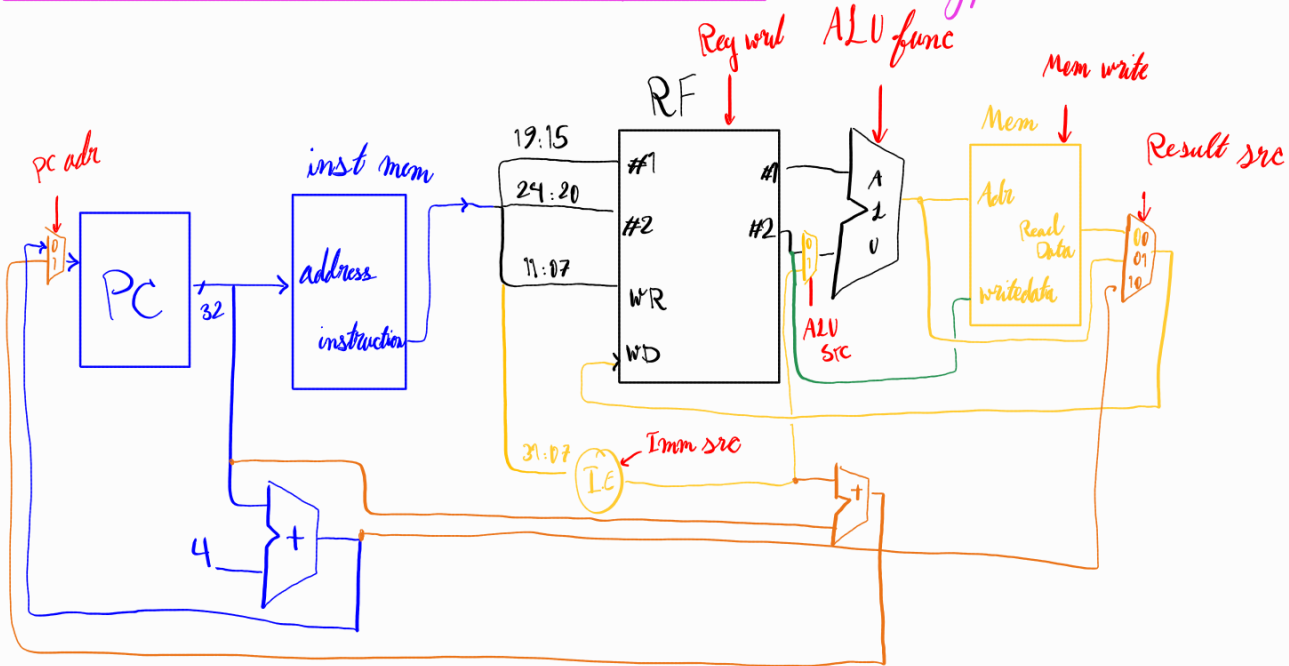
11:7

6:0

Imm[12], Imm[10:5]

r_tr_sf₃Imm, Imm
[7:7] [14]OP
Code

B-Type

اضافہ کریں دستور
bne, beq

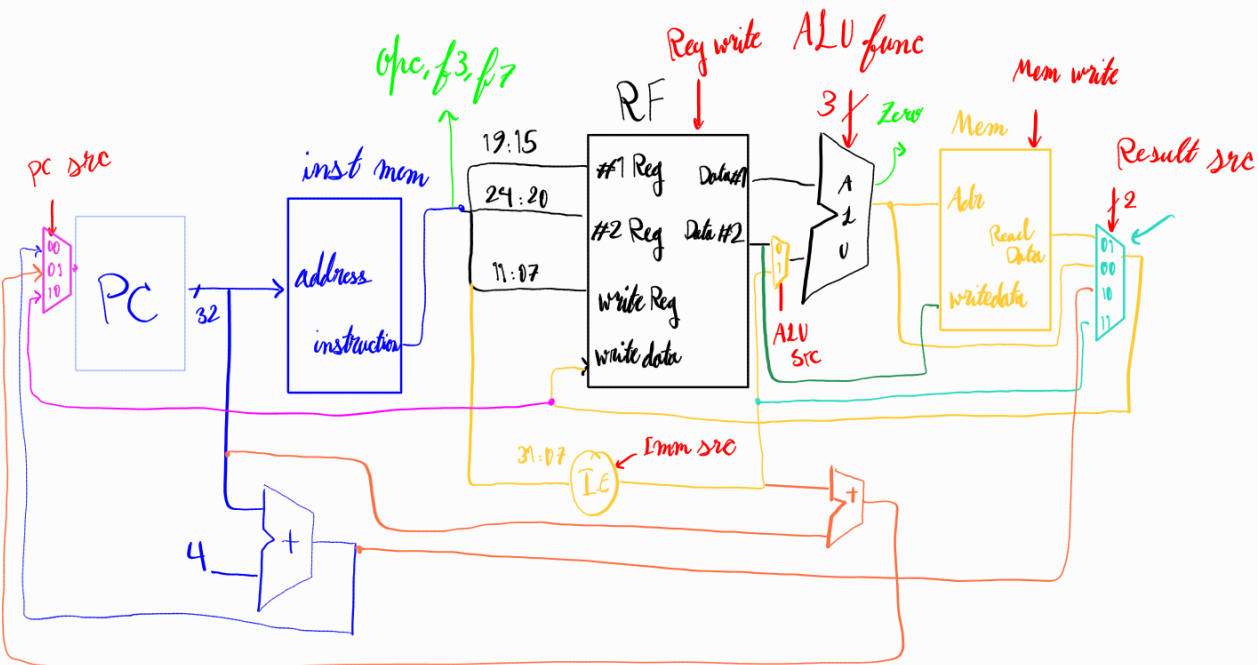
31:12

11:7

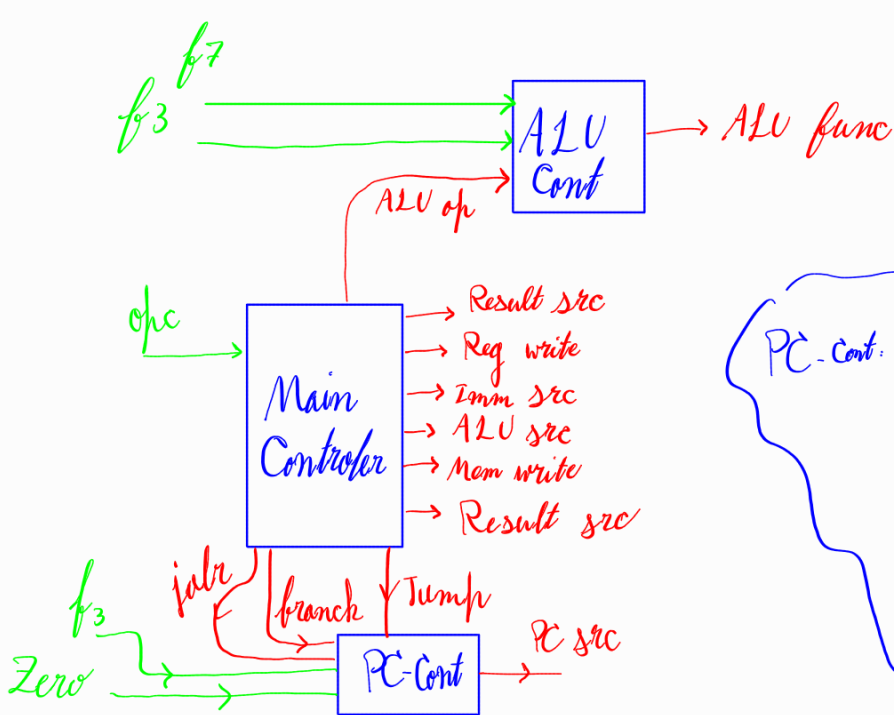
6:0

Imm[31:12]

rd

OP
CodeU-Type دستور
اضافہ کریں
lui r_d, imm

طراحی واحد کنترل



PC-Cont:

```

if (jump) PC-src = 01
elif (branch):
    if (f3 == beq && !Zero)
        PC-src = 01
    if (f3 == bne && !Zero)
        PC-src = 01
elif (jump):
    PC-src = 10
    
```

ALU cont:

ALU op	f3	f7	ALU func
00	X	X	000
01	X	X	001
10			000 001 010 011 100 101
11	addi ori ...	X	

Imm src:

Imm src		Type
000	$\{\{20\{I[31]\}\}, I[31:20]\}$	I-Type
001	$\{\{20\{I[31]\}\}, I[31:25], I[11:7]\}$	S-Type
010	$\{\{19\{I[31]\}\}, I[31], I[7], I[30:25], I[11:8], 1'b0\}$	B-Type
011	$\{\{12\{I[31]\}\}, I[19:12], I[20], I[30:21], 1'b0\}$	J-Type
100	$\{I[31:12], 12\{1'b0\}\}$	U-Type

Main Controller

opc	Inst	Reg write	Imm src	ALU src	mem write	Result src	branch	ALU op	Jump	jalu
51	R-T	1	XXX	0	0	01	0	10	0	0
3	lw	1	000	1	0	01	0	00	0	0
19	I-T	1	000	1	0	00	0	11	0	0
sw 35	S-T	0	001	1	1	XX	0	00	0	0
jal 111	J-T	1	011	X	0	10	0	XX	1	0
99	B-T	0	010	0	0	XX	1	01	0	0
lui 55	U-T	1	100	X	0	11	0	XX	0	0
103	jalr	1	000	1	0	00	0	00	0	1

* Assembly Code:

$\xrightarrow{s1}$
 $\xrightarrow{s0}$
 $\xrightarrow{s2}$
 int m = A[0];
 for (int i=0; i < 20; i++)
 if (m < A[i])
 m = A[i];

Jalr rd, rs, آدرس به بی → $\left\{ \begin{array}{l} rd \leftarrow pc + 4 \\ pc \leftarrow rs + \text{آدرس به بی} \end{array} \right.$

```

1  main:
2      addi s0, zero, 0 # 0 is the start of array
3      lw s1, 0(s0)
4      addi s2, zero, 0
5      addi s0, s0, 4
6  loop:
7      slti t0, s2, 10
8      beq t0, zero, end_loop
9      lw t0, 0(s0)
10     slt t1, s1, t0
11     bne t1, zero, else
12     add s1, t0, zero
13  else:
14     addi s2, s2, 1
15     addi s0, s0, 4
16     jal loop
17  end_loop:
18     addi s4, zero, 0
    
```

پاسخ نهایی $s1$ خواهد بود

